CHINTALAPUDI SREE CHARAN

AI/ML Engineer | Generative AI Engineer | Applied AI Engineer

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Summary

AI/ML Engineer who ships production-grade LLM systems: agentic analytics, grounded retrieval, and fine-tuned models, underpinned by distributed data engineering. Created a product that interrogates 10K+ comments per video, cross-checking every claim in code against its source material.

Experience

Freelance | Independent Contractor **Jun 2025 – Present**

AI & Backend Engineer / Sighnal – AI Content Strategist for YouTube Creators *Remote, India*

- Engineered an end-to-end **agentic/GenAI** system that scrapes every comment on a YouTube channel (10K+ threads/video, no API key) and distils it into audience intent, **BERTopic** topic clusters, content gaps, and a prioritized strategy backlog, on **FastAPI**, **Celery**, and **MongoDB**.
- Model responses risked citing outdated or unsubstantiated sources; devised a **grounded chatbot** where the LLM merely proposes claims and code validates each citation, lifting representative evidence from **15% to 52%** and eliminating stale stored-analysis answers (11% to 0%).
- Hardened large-batch **LLM classification** with AIMD concurrency, incremental persistence, and spaced auto-retries; a **12,910-comment** run improved from **574 permanently failed comments to 0 failed**, with Redis connection errors falling from **20+ to 0**. **SimHash** near-duplicate filtering trims a further **5–15%** of tokens.
- Implemented a self-healing **Celery watchdog** that detects and remedies 10 failure modes, backed by distributed locks and concurrency gates for fault-tolerant execution.
- Added **JWT**/RBAC and plan-tier enforcement, a **300+ test** suite, **Docker Compose** deployment, and a **Next.js** dashboard.

TCS Digital **Dec 2025 – Present**

Data Engineer *India*

- Architected multi-stage ETL workflows on **Azure Databricks** and **PySpark**, ingesting enterprise data into **Delta Lake** with schema enforcement and ACID guarantees, forming the feature layer that feeds downstream ML training and inference.
- Automated multi-source validation and standardization in Databricks notebooks, and tuned Spark execution (partitioning, broadcast joins, caching) to relieve bottlenecks in live AI workloads.
- Received the **Special Initiative Award** at **TCS Pune** for outperforming the entire training cohort in **HLD** and **LLD**.

AISMARTLIVE Solutions **Mar 2025 – Dec 2025**

Junior Data Scientist *Visakhapatnam, India*

- Designed **SenseMinds 360**, an IoT anomaly-detection platform that streams **172,800 sensor datapoints/day** from 30 sensors via **MQTT** into **InfluxDB** at 15-second resolution.
- Technical lead for a **12-member** team building predictive-maintenance models with **XGBoost** and Isolation Forest: **92% detection accuracy**, investigation time cut **87.5%** (8 hrs → 1 hr); **SHAP** explainability flagged failures **6–8 hours** early.

Technical Skills

Agentic AI & LLMs: LangGraph, LangChain, Langflow, AgentScope, Multi-Agent Systems, RAG, Vector Search, Prompt Engineering, PEFT, LoRA, Hugging Face, Groq, Claude, Gemini, Gemma

NLP, ML & CV: BERTopic, UMAP, HDBSCAN, sentence-transformers, XGBoost, Isolation Forest, SHAP, Scikit-learn, Pandas, YOLOv8, OpenCV

Data & Distributed Systems: Azure Databricks, PySpark, Delta Lake, ETL, Celery, Redis, MongoDB, PostgreSQL, InfluxDB, Weaviate, Azure AI Search

Backend, Deployment & Languages: FastAPI, Flask, Next.js, TypeScript, Streamlit, Docker, JWT/RBAC, Azure Container Registry, AWS ECR, Git, Python, SQL, Java, C

Projects

Multi-Agent Financial Analytics Platform | *LangGraph, PostgreSQL, Gemini* | GitHub

- Built a **LangGraph** multi-agent system turning natural-language queries into stock and risk analysis over live **PostgreSQL** data (5-stage workflow: intent, entities, Text-to-SQL, validation, synthesis) with SQL-injection guardrails and TTL caching.

Legal Document Summarization | *Tiny LLaMA 1.1B, Gemma 2-2B, PEFT, Databricks* | GitHub

- Co-developed a two-stage **PEFT** adaptation (Tiny LLaMA 1.1B → Gemma 2-2B) with extractive/abstractive summarization; **ROUGE-L 0.4373**, **0.8449 cosine similarity**, beating TextRank and BERTSUM; published in *IJIRT* (Mar 2025).

Education

Raghu Engineering College **Oct 2021 – May 2025**

B.Tech in Data Science | CGPA: ***8.8/10.0*** *Visakhapatnam, India*

Certifications

Certifications: Machine Learning (Coursera, Andrew Ng) · Generative AI using LangChain (Krish Naik) · DSA (Smart Interviews) · Python (DataPro) · GenAI Workshop (JNTU-V)

Achievements

- **Smart Interviews DSA:** Ranked **159 / 35,900+** globally (**Top 0.5%**), **1st** college-wide. **TCS CodeVita Season 12:** cracked the contest, **Global Rank 5K**. CodeChef **3-Star (1651)**, LeetCode **Knight (Top 4%)**, **500+** problems solved.
- **Founder & Lead, Codex Community:** Scaled a **400+ member** student tech community; delivered **75+** projects.
- **Hackathons: TCS AI Hackathon – Champion Certificate**; **RTGS AI** 3rd Prize twice (Predictive Protection, 500+ participants; Healthcare, 280+ entrants); **HackAP** Top 10 of 200+.